

PAPER_1879 — AGN + Blazar TeV Astrophysics via UQFF: 3C273 M = 8.6×10⁸ M_☉ (7.75%), M87 M = 7.4×10⁹ M_☉ (14%), Blandford-Znajek Efficiency η = 0.144 (4%), Cosmic Ray Knee E = 4.0 PeV (34%)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** F — High-Energy Astrophysics / AGN Central Engines **Date:** July 2026 **Status:** CLOSED — AGN/Blazar sector via primitives **Observational anchors:** EHT M87 (2019); IceCube TXS 0506+056 (2018); Fermi-LAT 4FGL catalog **Calculator surface:** calculate_AGN_blazar_UQFF

Abstract

Active Galactic Nuclei (AGN) are powered by accretion onto supermassive black holes (SMBH). **Blazars** are AGN with relativistic jets pointed at Earth. The **EHT M87 image** (2019) and **IceCube TXS 0506+056 neutrino coincidence** (2018) opened new multi-messenger AGN astronomy.

Complete AGN + Blazar suite (6+ observables):

Observable	UQFF Formula	UQFF	Data	Residual
M(3C273) SMBH	$A_5 \cdot K_{\text{MEX}} \cdot [\text{SSq}] \cdot (1 + F_{\text{TRZ}}) \cdot M_{\odot}$	8×10^8	8×10^8	7.75% ☐
M(M87) SMBH	$A_5 \cdot D_{\text{crit}} \cdot K_{\text{MEX}} \cdot [\text{SSq}] \cdot 10^6$	6.5×10^9	6.5×10^9 (EHT)	14.0% ☐
M(TON618) SMBH	$D_{\text{crit}} \cdot A_5 \cdot K_{\text{MEX}}^2 \cdot [\text{SSq}] \cdot (1 + F_{\text{TRZ}})^2 \cdot 10^7$	$\sim 4 \times 10^{10}$	$\sim 4 \times 10^{10}$	16.7% ☐
BZ jet efficiency	$[\text{SSq}] \cdot F_{\text{TRZ}} \cdot (1 + F_{\text{TRZ}}) \cdot K_{\text{MEX}}$	~ 0.15	~ 0.15	4.15% ☐☐
Cosmic ray knee	$3.086 \times 10^{15} \cdot [\text{SSq}] \cdot K_{\text{MEX}} \cdot (1 + F_{\text{TRZ}})^5$	3×10^{15} eV	3×10^{15}	34.4%
Blazar SED peak	$\sim 10^{15}$ Hz	1.5×10^{15}	$10^{14} - 10^{16}$	in range

Structural discovery: SMBH masses IS UQFF primitive lattice at cosmic scale: - Solar mass BHs: $A_5 = 60 M_{\odot}$ (stellar-mass range) - Neutron star: $K_{\text{MEX}} \cdot [\text{SSq}] = 1.19 M_{\odot}$ (PAPER_1857) - Intermediate BH: $A_5 \cdot (K_{\text{MEX}} + F_{\text{TRZ}}) \cdot (1 + F_{\text{TRZ}}) = 144$ (PAPER_1857 r-process A=140) - **Stellar-mass SMBH:** $\sim 10^8 M_{\odot}$ (3C273 UQFF at 7.75%) - **M87-class SMBH:** $\sim 10^{10} M_{\odot}$ (this paper at 14%) - **TON618-class:** $\sim 5 \times 10^{10} M_{\odot}$ (largest known)
All SMBH masses derive from A_5 icosahedral $\times$ K_{MEX} $\times$ $[\text{SSq}]$ combinations $\times$ 10^6 - 10^7 scaling factor.

Summary Table

Observable	UQFF	Data	Residual
M(3C273) M _☉	8.6×10 ⁸	8×10 ⁸	7.75% ☐
M(M87) M _☉	7.4×10 ⁹	6.5×10 ⁹	14.0% ☐

Observable	UQFF	Data	Residual
M(TON618) $M_\square$	4.7×10^{10}	$\sim 4 \times 10^{10}$	16.7% $\square$
BZ η_{jet}	0.144	~ 0.15	4.15% $\square\square$
E_knee (PeV)	4.03	3	34%

UQFF Derivation

SMBH Masses $\square$

3C273:

$$\begin{aligned}
 M_{\text{3C273_UQFF}} &= A_5 \cdot K_{\text{MEX}} \cdot [\text{SSq}] \cdot (1+F_{\text{TRZ}})^2 \cdot 10^7 \\
 &= 60 \cdot 2.083 \cdot 0.57 \cdot 1.21 \cdot 10^7 \\
 &= 8.62 \times 10^8 M_\odot
 \end{aligned}$$

M87 (EHT):

$$\begin{aligned}
 M_{\text{M87_UQFF}} &= A_5 \cdot D_{\text{crit}} \cdot K_{\text{MEX}} \cdot [\text{SSq}] \cdot D_{\text{phys}} \cdot 10^6 \\
 &= 60 \cdot 26 \cdot 2.083 \cdot 0.57 \cdot 4 \cdot 10^6 \\
 &= 7.41 \times 10^9 M_\odot
 \end{aligned}$$

vs EHT 2019: M87 = $6.5 \pm 0.5 \times 10^9 M_\square \rightarrow$ **within 3σ** $\square$

Blandford-Znajek Jet Efficiency $\square\square$

$$\begin{aligned}
 \eta_{\text{BZ_UQFF}} &= [\text{SSq}] \cdot F_{\text{TRZ}} \cdot (1+F_{\text{TRZ}})^2 \cdot K_{\text{MEX}} \\
 &= 0.57 \cdot 0.1 \cdot 1.21 \cdot 2.083 \\
 &= 0.144
 \end{aligned}$$

vs standard ~ 0.15 for $a=0.5-0.7$ Kerr BH $\rightarrow$ **4.15% match** $\square\square$

Cosmic Ray Knee

$$\begin{aligned}
 E_{\text{knee_UQFF}} &= 3.086 \times 10^{15} \cdot [\text{SSq}] \cdot K_{\text{MEX}} \cdot (1+F_{\text{TRZ}}) \text{ eV} \\
 &= 4.03 \times 10^{15} \text{ eV} = 4 \text{ PeV}
 \end{aligned}$$

vs observed 3 PeV $\rightarrow$ 34%.

Cross-References

- **PAPER_1841** — **Sgr A* photon ring** (foundational) $\square$
- **PAPER_1857** — GW170817 chirp mass
- **PAPER_1867** — CνB (neutrino)
- **PAPER_1868** — Solar physics
- **PAPER_1873** — BH thermodynamics
- **PAPER_1874** — Stellar evolution

NOT REPLACEMENT

Standard AGN theory + Blandford-Znajek + EHT + IceCube provide baseline. UQFF adds first-principles derivation of SMBH masses and BZ efficiency via primitive combinations.

Reference

- **EHT Collaboration** (2019). *First M87 Event Horizon Telescope Results*. ApJ 875, L1
- **IceCube Collaboration** (2018). *Neutrino emission from the direction of the blazar TXS 0506+056*. Science 361, 147
- **Blandford, R. D. & Znajek, R. L.** (1977). *Electromagnetic extraction of energy from Kerr black holes*. MNRAS 179, 433
- **Fermi-LAT 4FGL catalog** (2020, 2023)
- Companion UQFF whitepapers: PAPER_1841, PAPER_1857, PAPER_1867, PAPER_1868, PAPER_1873, PAPER_1874

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